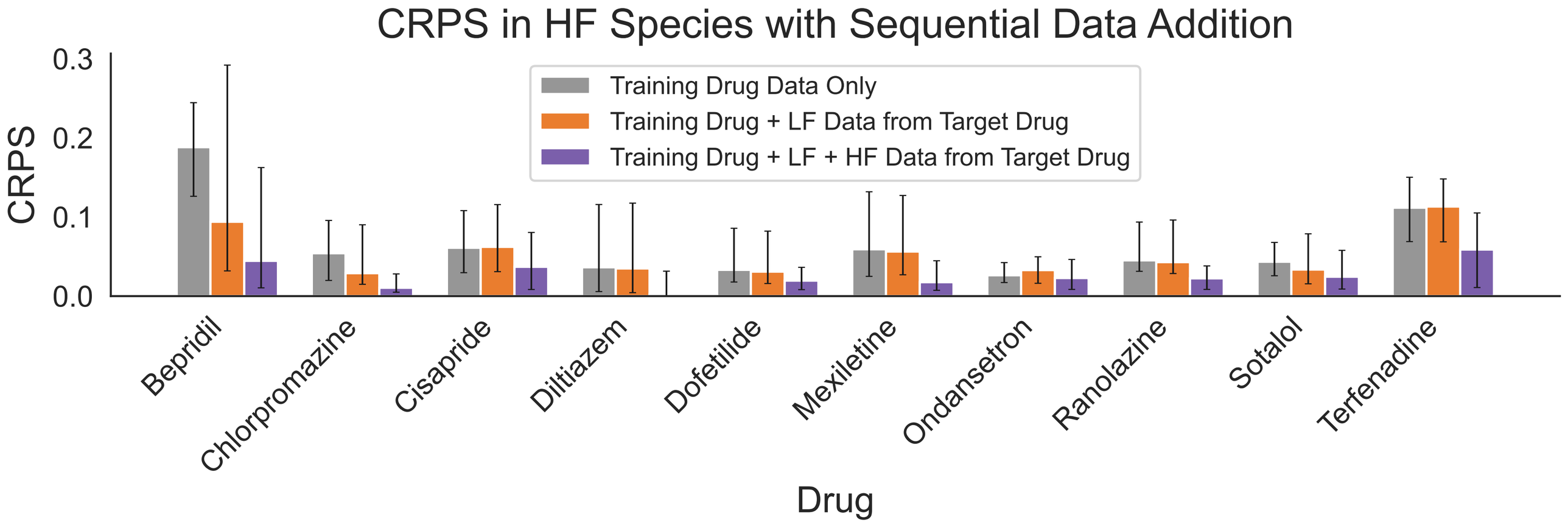
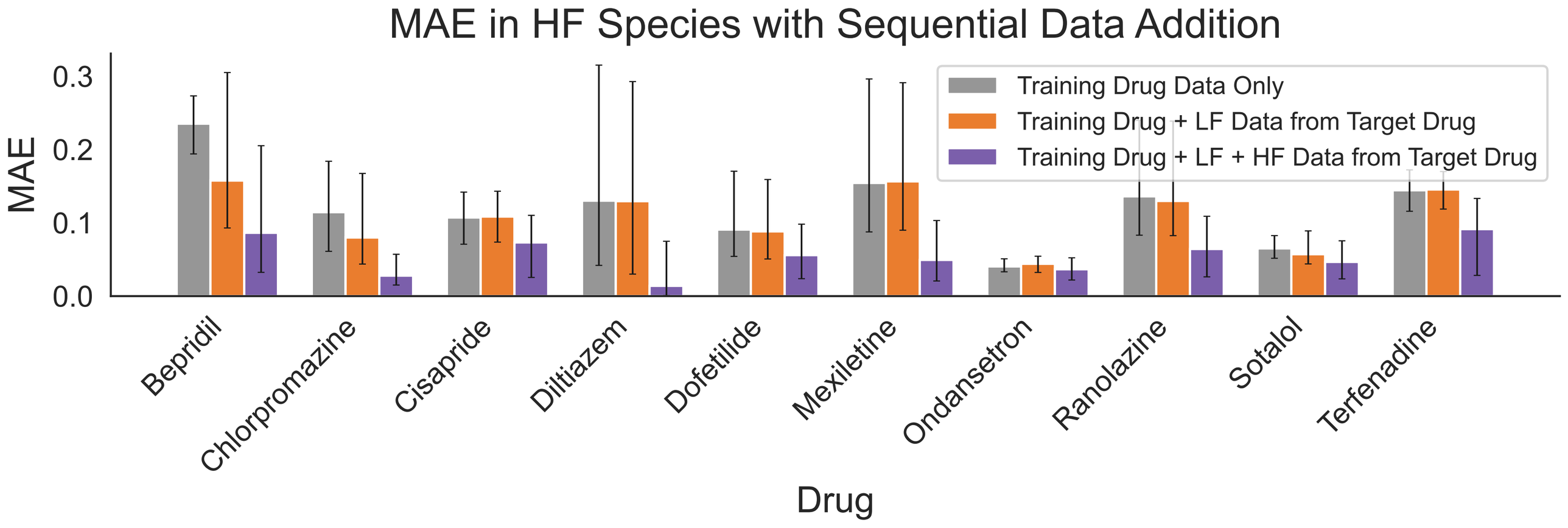
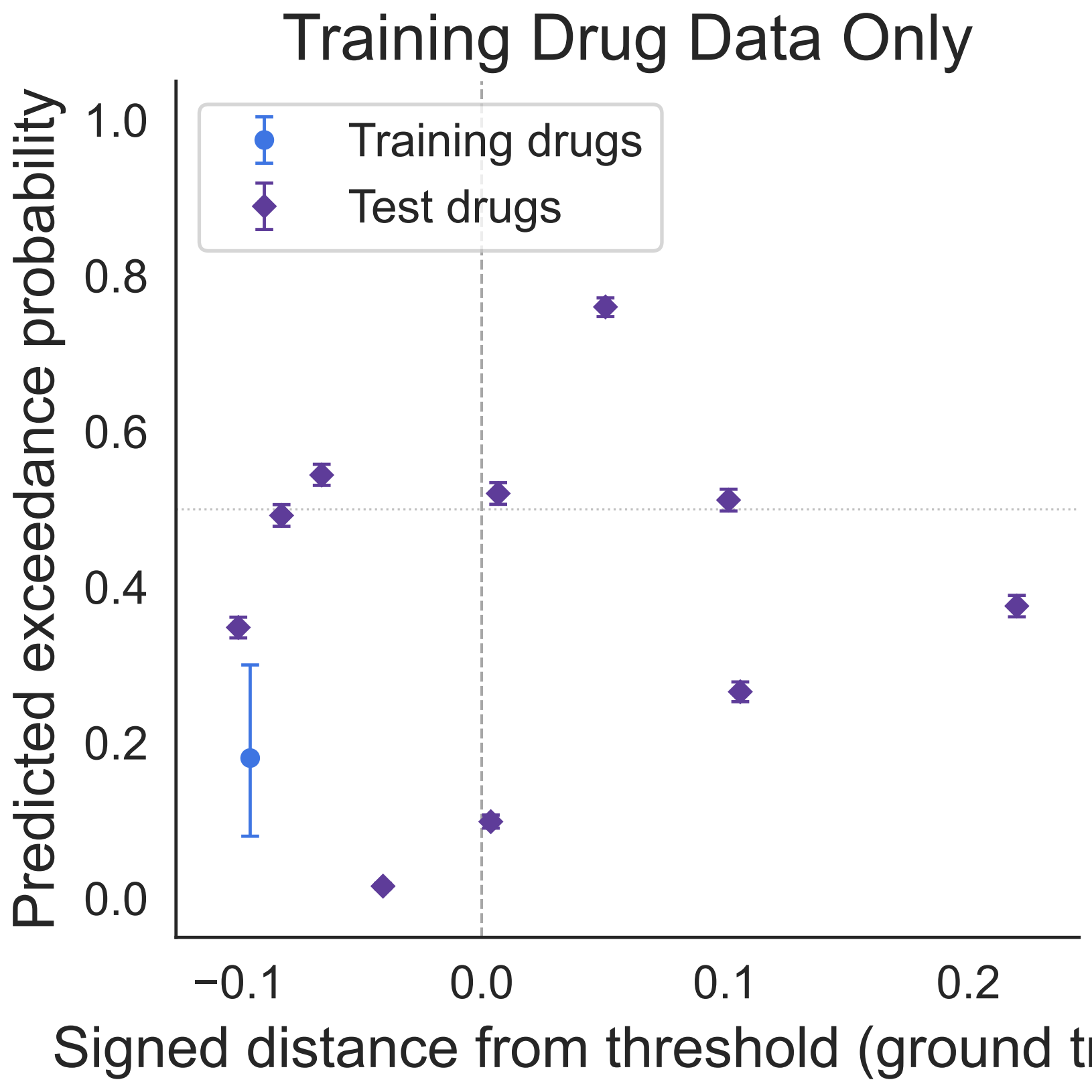

CRPS in HF Species with Sequential Data Addition
CRPS
0.0
0.1
0.2
0.3
Training Drug Data Only
Training Drug + LF Data from Target Drug
Training Drug + LF + HF Data from Target Drug
Bepridil
Chlorpromazine
Cisapride
Diltiazem
Dofetilide
Mexiletine
Ondansetron
Ranolazine
Sotalol
Terfenadine
Drug

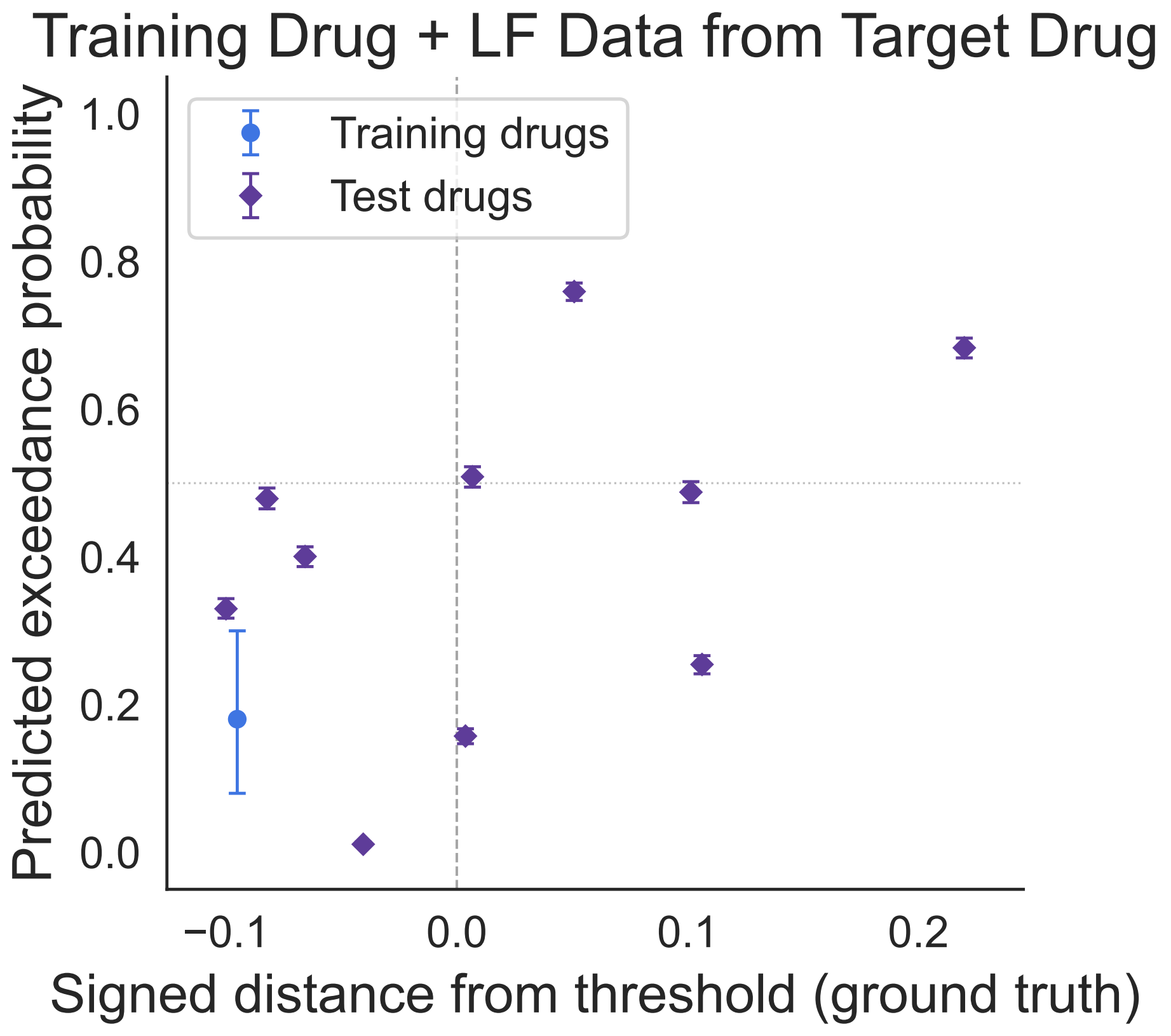

MAE in HF Species with Sequential Data Addition
MAE
0.0
0.1
0.2
0.3
Training Drug Data Only
Training Drug + LF Data from Target Drug
Training Drug + LF + HF Data from Target Drug
Bepridil
Chlorpromazine
Cisapride
Diltiazem
Dofetilide
Mexiletine
Ondansetron
Ranolazine
Sotalol
Terfenadine
Drug

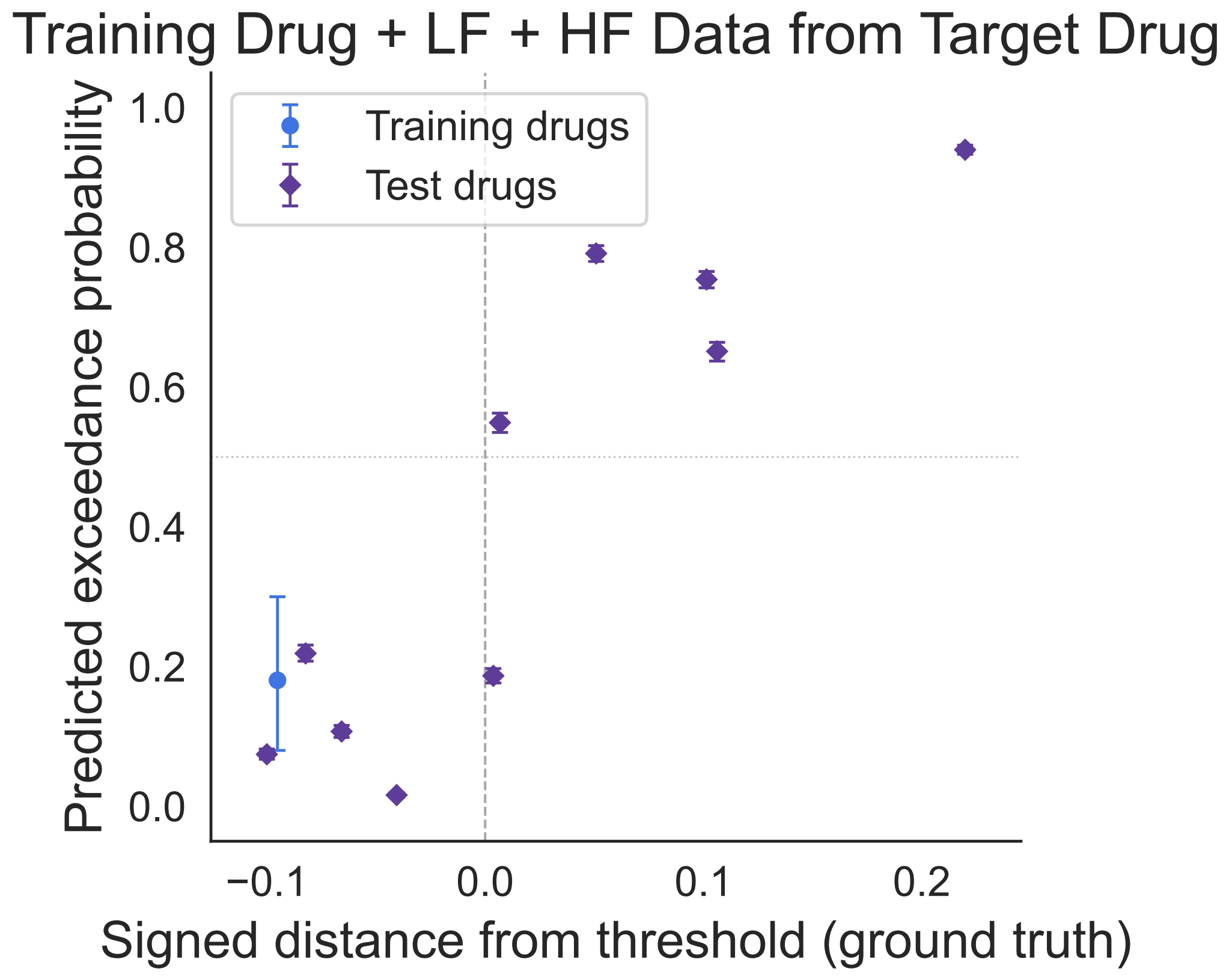

Sharpness ($\frac{1}{2}\,\mathbb{E}|X - X'|$) in HF Species with Sequential Data Addition
Sharpness ($\frac{1}{2}\,\mathbb{E}|X - X'|$)
0.0
0.1
0.2
Training Drug Data Only
Training Drug + LF Data from Target Drug
Training Drug + LF + HF Data from Target Drug
Bepridil
Chlorpromazine
Cisapride
Diltiazem
Dofetilide
Mexiletine
Ondansetron
Ranolazine
Sotalol
Terfenadine
Drug

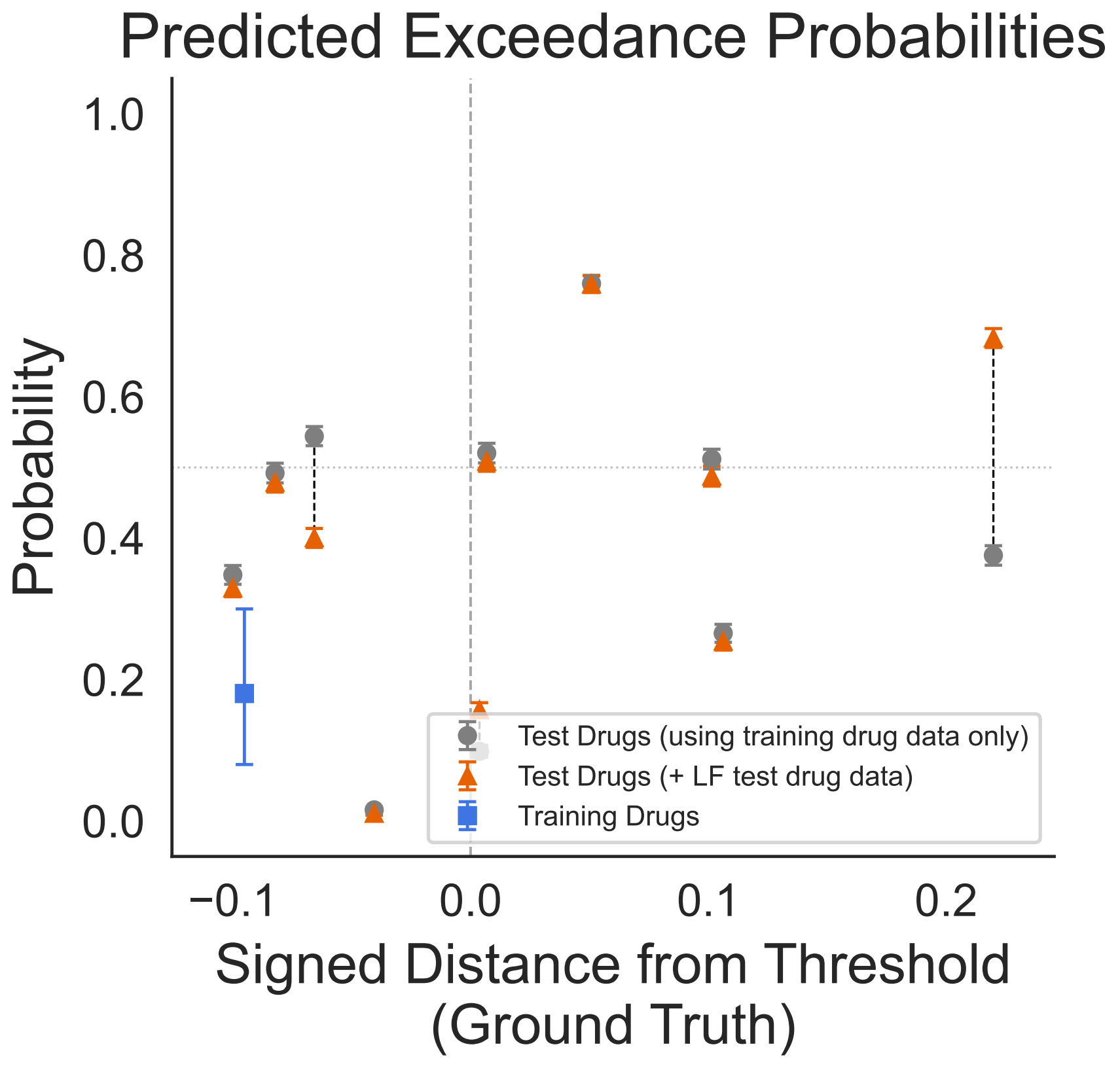
Exceedance Probability per Drug and Scenario (Median with 95% CI)

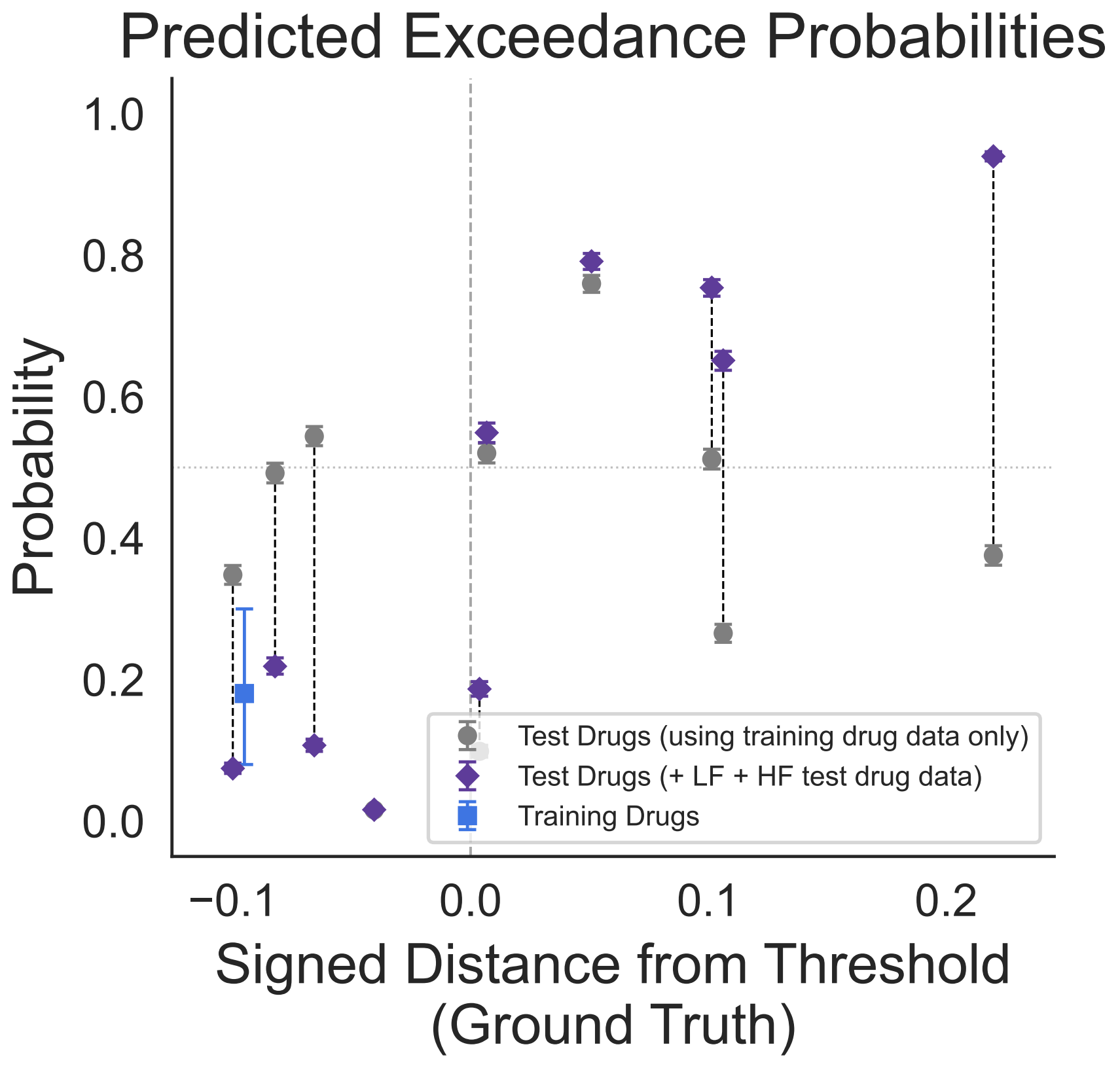
Drug	Training Drug Data Only	Training Drug + LF Data from Target Drug	Training Drug + LF + HF Data from Target Drug
Bepridil	0.376 (0.362, 0.389)	0.683 (0.670, 0.696)	0.940 (0.933, 0.946)
Chlorpromazine	0.544 (0.531, 0.558)	0.401 (0.387, 0.414)	0.107 (0.099, 0.116)
Cisapride	0.512 (0.498, 0.526)	0.488 (0.474, 0.502)	0.754 (0.742, 0.766)
Diltiazem	0.348 (0.335, 0.361)	0.330 (0.317, 0.344)	0.074 (0.067, 0.082)
Dofetilide	0.760 (0.748, 0.772)	0.759 (0.747, 0.771)	0.791 (0.780, 0.803)
Mexiletine	0.492 (0.478, 0.506)	0.479 (0.465, 0.493)	0.219 (0.208, 0.231)
Ondansetron	0.016 (0.012, 0.020)	0.011 (0.008, 0.014)	0.016 (0.013, 0.019)
Ranolazine	0.520 (0.506, 0.534)	0.509 (0.495, 0.522)	0.549 (0.535, 0.563)
Sotalol	0.099 (0.090, 0.107)	0.158 (0.147, 0.167)	0.187 (0.177, 0.197)
Terfenadine	0.266 (0.253, 0.278)	0.254 (0.242, 0.266)	0.651 (0.637, 0.664)

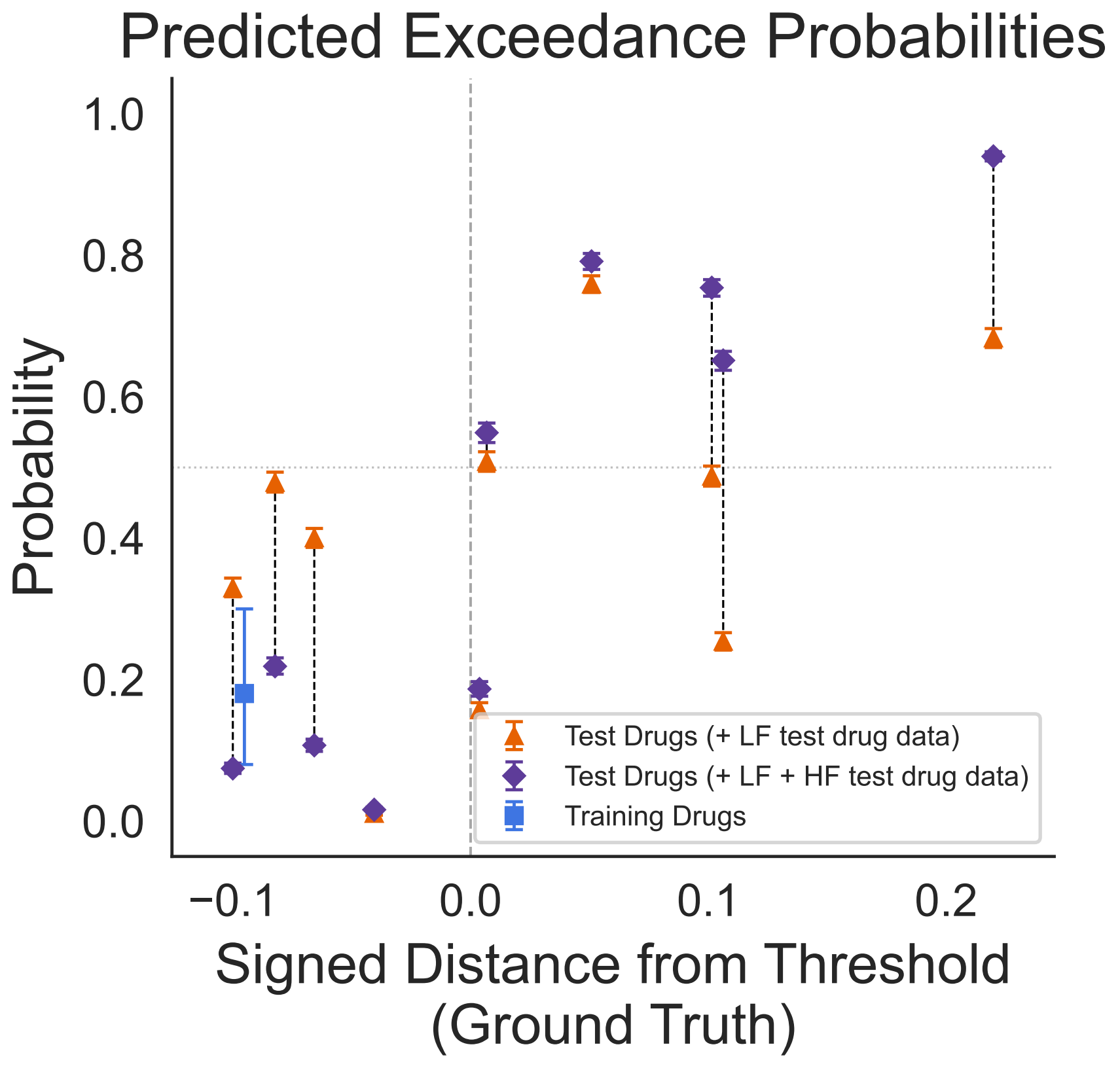

Training Drug Data Only
Training drugs
Test drugs
Predicted exceedance probability
1.0
0.8
0.6
0.4
0.2
0.0
−0.1
0.0
0.1
0.2
Signed distance from threshold (ground t

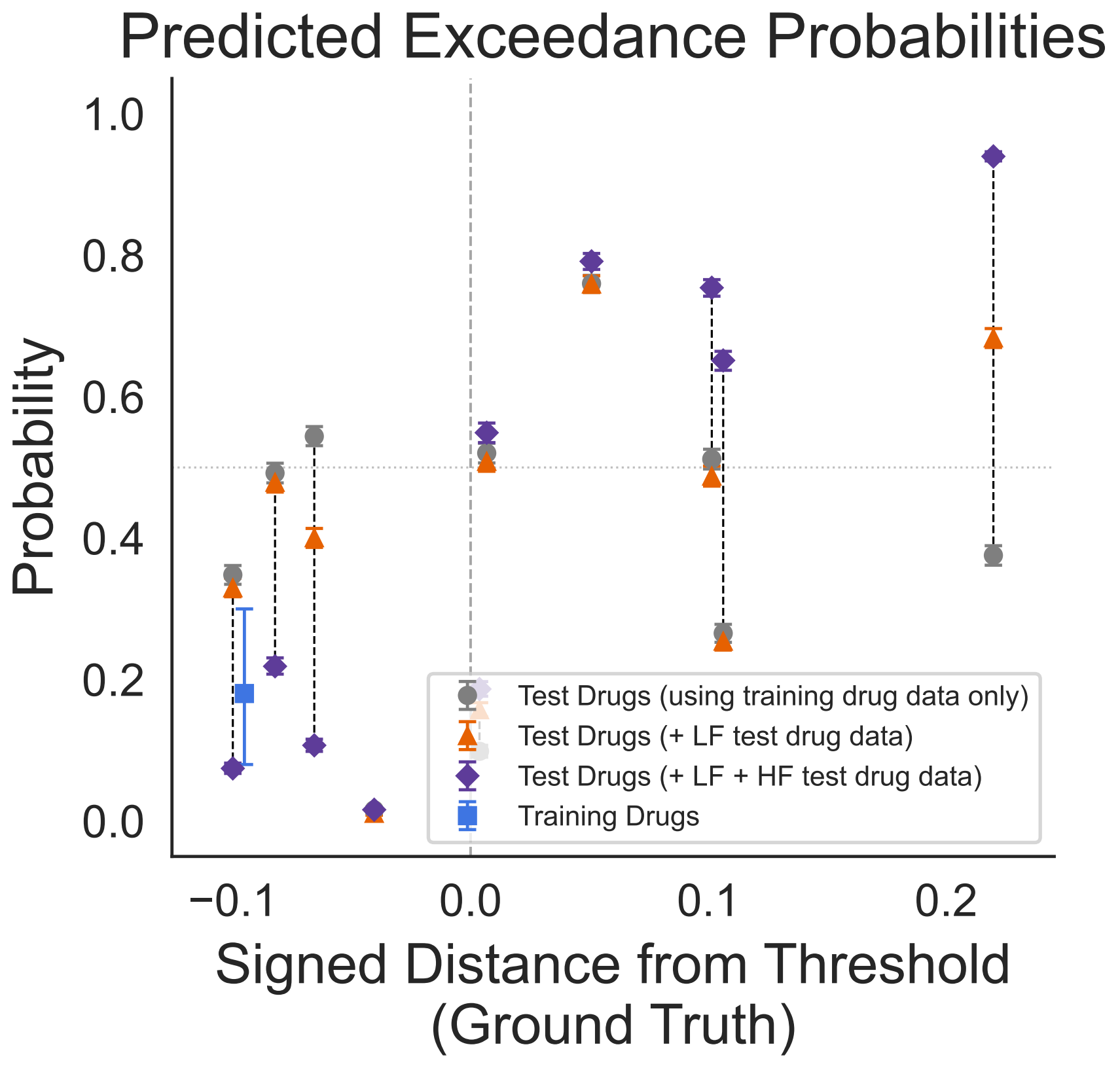

Training Drug + LF Data from Target Drug
Training drugs
Test drugs
Predicted exceedance probability
1.0
0.8
0.6
0.4
0.2
0.0
−0.1
0.0
0.1
0.2
Signed distance from threshold (ground truth)


Training Drug + LF + HF Data from Target Drug
Predicted exceedance probability
1.0
0.8
0.6
0.4
0.2
0.0
Training drugs
Test drugs
−0.1
0.0
0.1
0.2
Signed distance from threshold (ground truth)


Predicted Exceedance Probabilities
Probability
1.0
0.8
0.6
0.4
0.2
0.0
−0.1
0.0
0.1
0.2
Signed Distance from Threshold
(Ground Truth)
Test Drugs (using training drug data only)
Test Drugs (+ LF test drug data)
Training Drugs


Predicted Exceedance Probabilities
Probability
1.0
0.8
0.6
0.4
0.2
0.0
−0.1
0.0
0.1
0.2
Signed Distance from Threshold
(Ground Truth)
Test Drugs (using training drug data only)
Test Drugs (+ LF + HF test drug data)
Training Drugs


Predicted Exceedance Probabilities
Probability
1.0
0.8
0.6
0.4
0.2
0.0
−0.1
0.0
0.1
0.2
Signed Distance from Threshold
(Ground Truth)
Test Drugs (+ LF test drug data)
Test Drugs (+ LF + HF test drug data)
Training Drugs


Predicted Exceedance Probabilities
Probability
1.0
0.8
0.6
0.4
0.2
0.0
−0.1
0.0
0.1
0.2
Signed Distance from Threshold
(Ground Truth)
Test Drugs (using training drug data only)
Test Drugs (+ LF test drug data)
Test Drugs (+ LF + HF test drug data)
Training Drugs

Training drugs (in-sample) – Tacrine (drug-level)

